

Play Pulse Studios Online Game Data Analysis

COMP1013

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1 Player Engagement Analysis By Experience Level

1.1 Experience Group

In order to analyse player behaviour across different experience levels, we will run the code to group them into:

- Veteran: before 2023
- Intermediate: 2023
- New: after 2023

```
players = players %>%
  mutate(
    # Convert signup_date to a proper date format
    signup_date = as.Date(signup_date),
    # First ifelse() checks if player signed up before 2023 -> "Veteran"
    # If not, the second ifelse() checks if they signed up in 2023 -> "Intermediate"
    # If neither condition is met, it means they signed up after 2023 -> "New"
    experience_group = ifelse(year(signup_date) < 2023, "Veteran",
                              ifelse(year(signup_date) == 2023, "Intermediate",
                                      "New")))
    # Then is stored into the experience_group variable
```

1.2 Number Of Players

```
# Count the number of players in each experience group
num_players = players %>%
  # The code groups the rows by experience_group
  group_by(experience_group) %>%
  # Function n() counts the number of rows in each group, which gives us the
  # number of players per group
  summarise(number_of_players = n())
```

1.2.1 Using kableExtra To Tabulate The Number Of Players Per Group

```
num_players %>%
  kbl(
    # Rename the columns
    col.names = c("Experience Group", "Number of Players"),
    # # Left align the first column and centre align the second
    align      = c("l", "c")
  ) %>%
  kable_styling(
    # "striped" to change row colours
    # "bordered" adds lines around every cell
    bootstrap_options = c("striped", "bordered"),
    full_width = FALSE, # table is only as wide as it needs to be
```

```

    position = "left" # aligns table to the left side of page
  ) %>%
  # Styles the header row and colour of words
  row_spec(0, bold = TRUE, background = "#333333", color = "white") %>%
  column_spec(2, color = "black")

```

Experience Group	Number of Players
Intermediate	1851
New	1363
Veteran	1786

1.2.2 Findings For Number Of Players In Each Group

The data shows that across the three experience groups, the Intermediate group is the largest, with **1,851 players**. Veterans follow with **1,786 players**, while New players make up the smallest group at **1,363 players**.

Intermediate players account for the biggest portion of the player base, which suggests that 2023 was a particularly strong year for new sign-ups on the platform. Veterans come in as the second largest group, indicating a solid and stable community of long-term players who joined before 2023.

Even though New players form the smallest group, their total of 1,363 players is still quite significant. This shows that the platform continues to attract fresh users even after 2023.

Taken together, these figures point to a healthy and steadily growing play base for Play Pulse Studios, supported by a strong core of experienced players and a consistent stream of new arrivals over time.

1.3 Average Play Time By Experience Group

```

# The function left_join keeps all sessions and matches player info where available
# It is stored in the sessions_joined variable
sessions_joined = sessions %>%
  left_join(players %>% select(c(player_id, experience_group)), by = "player_id")

# cat() prints the value for sessions that could not be matched to a player (NA values)
cat("The NA values for sessions that matched to a player is",
    sum(is.na(sessions_joined$experience_group)))

```

The NA values for sessions that matched to a player is 0

Now that we have determined the NA value is 0, we do not need to filter it out again.

```

avg_playtime_group = sessions_joined %>%
  # Groups by each experience group
  group_by(experience_group) %>%
  # Removes any NA values then calculate the mean of the play time minutes in each group
  summarise(avg_play_time_mins = mean(play_time_minutes, na.rm = TRUE))

```

1.3.1 Using 'ggplot2' To Plot The Average Play Time By Experience Group

```
# Use avg_playtime_group to create a bar chart
ggplot(avg_playtime_group,
      aes(x = experience_group,
          y = avg_play_time_mins,
          fill = experience_group)) +

# Draws the bars
geom_col(width = 0.5) +

# Labels and small adjustments
geom_text(
  aes(label = paste0(avg_play_time_mins, " min")),
  vjust = -0.5,
  size = 4
) +

# Assign colours to the bar of each experience group
scale_fill_manual(values = c(
  "Veteran" = "#8EA58C",
  "Intermediate" = "#BFCFBB",
  "New" = "#344C3D"
)) +

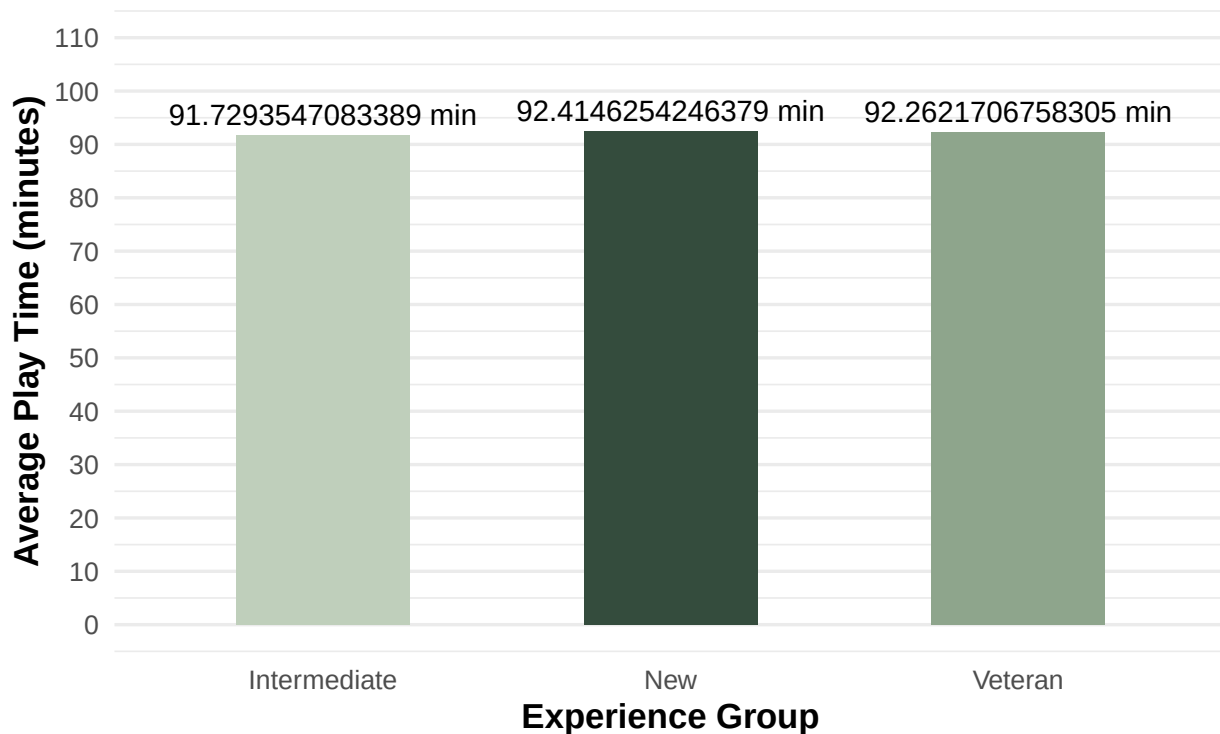
# Leaves space above bars for labels
scale_y_continuous(
  limits = c(0, 110),
  breaks = seq(0, 110, 10)
) +

# Adds titles and axis labels
labs(
  title = "Average Play Time By Player Experience Group",
  subtitle = "Based on player signup date",
  x = "Experience Group",
  y = "Average Play Time (minutes)",
) +

# Applies themes to make it look nice
theme_minimal(base_size = 13) +
theme(
  legend.position = "none",
  plot.title = element_text(face = "bold", size = 14),
  plot.subtitle = element_text(colour = "#5D6D7E"),
  axis.title = element_text(face = "bold"),
  panel.grid.major.x = element_blank()
)
```

Average Play Time By Player Experience Group

Based on player signup date



1.3.2 Findings For Average Play Time In Each Group

The data shows that players spend nearly the same amount of time in each session, regardless of how long they have been on the platform.

New players tend to stay the longest, with an average session time of **92.41 minutes**. Veterans are very close behind at **92.26 minutes**, while Intermediate players average **91.73 minutes**. The difference between the highest and lowest groups is just **0.68 minutes**, which is extremely small.

This level of consistency suggests that once players join the platform, their session length are not largely affected by their level of experience. Engagement appears to stay steady across all groups.

The slightly longer sessions among New players could be because they are still getting familiar with the platform. They are likely exploring different features and trying out various games. In contrast, Veterans may spend their time more efficiently since they already know what they enjoy and tend to focus on those activities.

1.4 Average Score Of Each Experience Group

```
# Using sessions_joined from earlier so we can calculate the average score per group
avg_score_group = sessions_joined %>%
  # Filters out if there are any sessions that have NA in the experience_group variable
  # !is.na find rows where group is NOT missing
```

```
filter(!is.na(experience_group)) %>%
# Group by experience_group so the calculation is done per group
group_by(experience_group) %>%
# Removes any NA values then calculates the average score per group
summarise(avg_score_per_group = mean(score, na.rm = TRUE))
```

1.4.1 Using kableExtra To Tabulate The Average Score Per Group

```
avg_score_group %>%
  kbl(
    # Rename the columns
    col.names = c("Experience Group", "Average Score"),
    # Left align the first column and centre align the second
    align      = c("l", "c")
  ) %>%
  kable_styling(
    # "striped" to change row colours
    # "bordered" adds lines around every cell
    bootstrap_options = c("striped", "bordered"),
    full_width = FALSE, # table only as wide as it needs to be
    position = "left" # align table to the left of the page
  ) %>%
  # Styles the header row and colour of words
  row_spec(0, bold = TRUE, background = "#333333", color = "white") %>%
  column_spec(2, bold = TRUE, color = "black")
```

Experience Group	Average Score
Intermediate	498.2044
New	495.9630
Veteran	501.0407

1.4.2 Findings For Average Scores Per Group

The data on average scores across the three experience groups shows that Veterans achieved the highest average score at **501.04**, followed by Intermediate players at **498.20**, and New players at **495.96**.

It is not surprising that Veterans come out on top, as their longer time on the platform gives them more opportunity to build strong skills and refine their strategies. With the most experience, they are more familiar with the games and can perform at a higher level.

Intermediate players fall in the middle with an average score of **498.20**, which shows their moderate level of experience. They have had enough time to develop some skill but have not yet reached the same level of proficiency as Veterans.

New players recorded the lowest average score at **495.96**, which makes sense since they have only recently joined and are still learning the mechanics. However, the gap between New players and Veterans is just **5.08 points**, which is surprisingly small. This suggests that new players are picking things up quickly and adapting well to the games.

Looking at the results as a whole, the scores across all three groups are very close. This indicates that the games on the platform are accessible and enjoyable for players at all experience levels, allowing new users to stay competitive while still rewarding more experienced players.

2 Game Genre Analysis

2.1 Average Play Time For Each Genre

```
# The function left_join keeps all sessions and matches player info where available
# It is stored in the sessions_games variable
sessions_games = sessions %>%
  left_join(
    games %>% select(game_id, genre), by = "game_id")

# cat() prints the value for sessions that could not be matched to a game (NA values)
cat("The NA values for sessions that matched to a game is",
    sum(is.na(sessions_games$genre)))
```

The NA values for sessions that matched to a game is 0

Now that we have determined the NA value is 0, we do not need to filter it out again.

```
avg_playtime_genre = sessions_games %>%
  # Groups by each genre
  group_by(genre) %>%
  # Removes any NA values then calculates the mean of the play time minutes in each genre
  summarise(avg_play_time_mins2 = mean(play_time_minutes, na.rm = TRUE))
```

2.1.1 Using kableExtra To Tabulate The Average Playtime In Each Genre

```
avg_playtime_genre %>%
  kbl(
    # Rename the columns
    col.names = c("Genre", "Average Play Time (min)"),
    # Left align the first column and centre align the second
    align = c("l", "c")
  ) %>%
  kable_styling(
    # "striped" to change row colours
    # "bordered" adds lines around every cell
    bootstrap_options = c("striped", "bordered"),
    full_width = FALSE, # table only as wide as it needs to be
    position = "left" # align table to the left side of the page
  ) %>%
  # Styles the header row and colour of words
  row_spec(0, bold = TRUE, background = "#333333", color = "white") %>%
  column_spec(2, bold = TRUE, color = "black")
```

Genre	Average Play Time (min)
Action	92.09816
Puzzle	91.67226
RPG	91.41153
Sports	92.58573
Strategy	92.88334

2.1.2 Findings For Average Play Time In Each Genre

The data on average play time across game genres shows that session length is quite consistent, with all genres averaging between **91 and 92 minutes** per session.

RPG games have the **highest average play time**, which makes sense given they are usually story driven and require longer sessions to progress in the game. Their complex and immersive story lines motivate players to stay engaged for longer periods.

On the other hand, Strategy games have the **lowest average play time**. This could be due to the way these games are structured, as players may prefer to complete shorter sessions in games. Since progress can usually be saved and continued later, there is less need for long, continuous play.

Action and Sports games sit in the middle, with very similar average session times. This may suggest that engagement levels maintain consistent session length regardless of the type of game being played, as players are spending a similar amount of time per session across these genres.

The small differences in play time indicate that the platform maintains a steady level of engagement across all game types. No matter what genre players choose, session lengths remain stable.

2.2 Average Score Per Genre

```
# Using sessions_games from earlier so we can calculate average score per genre
avg_score_genre = sessions_games %>%
  # Filters out if there are any sessions that have NA in the genre variable
  # !is.na find rows where genre is NOT missing
  filter(!is.na(genre)) %>%
  # The code groups by genre so the calculation is done per genre
  group_by(genre) %>%
  # Removes any NA values then calculates the average score per genre
  summarise(avg_score_per_genre = mean(score, na.rm = TRUE))
```

2.2.1 Using kableExtra To Tabulate The Average Score Per Genre

```
avg_score_genre %>%
  kbl(
    # Rename the columns
    col.names = c("Genre", "Average Score by Genre"),
    # Left align the first column and centre align the second
    align = c("l", "c")
  ) %>%
  kable_styling(
    # "striped" to change row colours
    # "bordered" adds lines around every cell
    bootstrap_options = c("striped", "bordered"),
    full_width = FALSE, # table only as wide as it needs to be
    position = "left" # align table to the left side of the page
  ) %>%
  # Styles the header row and colour of words
  row_spec(0, bold = TRUE, background = "#333333", color = "white") %>%
  column_spec(2, bold = TRUE, color = "black")
```

Genre	Average Score by Genre
Action	493.0569
Puzzle	496.5239
RPG	499.6938
Sports	503.3790
Strategy	500.8435

2.2.2 Findings For Average Score Per Genre

The data on average scores across game genres shows that RPG games have the **highest average score**, followed by Sports, Action, and Strategy games.

RPGs leading in average scores may suggest that players who pick this genre tend to perform better overall. This could be because RPGs often reward time and dedication. As seen in the playtime analysis, RPG players typically spend the longest time per session, and that extra time may help them achieve higher scores.

Strategy games, on the other hand, have the **lowest average score**. This may be due to the genre's complexity, as these games require careful planning, decision-making, and long-term thinking. Because of this higher difficulty, players may find it harder to consistently achieve high scores.

Action and Sports games sit in the middle, with very similar average scores. This suggests that both genres offer a comparable level of challenge and player performance.

Looking at the results, the differences in average scores across genres are quite small. This points to a fairly balanced scoring system on the platform, where players can achieve similar results regardless of the type of game they choose.

2.3 Number Of Sessions Per Genre

```
# Assigning num_sessions_genre variable to number of sessions per game
num_sessions_genre = sessions_games %>%
  # Filters out if there are any sessions that have NA in the genre variable
  # !is.na find rows where genre is NOT missing
  filter(!is.na(genre)) %>%
  # Groups by genre so the calculation is done per genre
  group_by(genre) %>%
  # Function n() counts the number of sessions per genre
  summarise(num_of_sessions_genre = n())
```

2.3.1 Using kableExtra To Tabulate The Number Of Sessions Per Genre

```
num_sessions_genre %>%
  kbl(
    # Rename the columns
    col.names = c("Genre", "Number of Sessions"),
    # Left align the first column and centre align the second
    align = c("l", "c")
  ) %>%
  kable_styling(
```

```
# "striped" to change row colours
# "bordered" adds lines around every cell
bootstrap_options = c("striped", "bordered"),
full_width = FALSE, # table only as wide as it needs to be
position = "left" # align table to the left side of the page
) %>%
# Styles the header row and colour of words
row_spec(0, bold = TRUE, background = "#333333", color = "white") %>%
column_spec(2, bold = TRUE, color = "black")
```

Genre	Number of Sessions
Action	3199
Puzzle	6371
RPG	2792
Sports	3995
Strategy	3643

2.3.2 Findings For Number Of Sessions Per Genre

The data of the number of sessions across game genres shows that some genres are played more frequently than others on the platform.

RPG games recorded the **highest number of sessions**, making them the most frequently played genre. This matches the earlier findings that RPGs also have the highest average play time, suggesting that players not only return to these games more often but also spend longer in each session. This shows RPGs as the most engaging genre on the platform.

In contrast, Strategy games have the **lowest number of sessions**, indicating they are played less often. This could reflect the more challenging aspects of the genre, as these games do often require careful thinking and a higher level of skill. Due to this fact, players may be less likely to come back as frequently compared to other genres.

Action and Sports games sit in between, with session counts that are quite close to each other. This suggests that both genres maintain a similar level of popularity and consistent player activity.

Altogether, these differences in session counts show that while all genres attract regular players, particularly RPGs, encourage more frequent gameplay. This insight could help Play Pulse Studios focus on expanding popular genres like RPGs.

2.4 Number Of Unique Players Per Genre